You must not read the question paper before the exam starts.
You must not write anything until the invigilator says that you may start.
Pay attention to the instructions given by the invigilators.
Read the instructions below **carefully** before the exam starts, **check** you know how you are required to complete your answer booklets or sheets.

You must ensure that the front of your answer books/sheets have been completed correctly **by the end of your exam finish time**, no further time will be given.

May/June 2025 **EC144 2024/5 A 800**

Approved calculators permitted
Answer booklet

University Of Reading

Mathematical Methods for Economists (EC144)

Three hours

Answer **all** questions.

1. Differentiate the following functions with respect to x .

(a) $y = (5x^3 - 3x^2 + 25x)^{\frac{1}{3}}$

(3 marks)

(b) $y = \ln\,(x^3 - \frac{5}{x} + 10)$

(3 marks)

(c) $5.6^2 = \frac{2}{y^2} - 2x^2 + y^3$ (Hint: y is an implicitly defined function of x.)

(4 marks)

2. Consider the two matrices below

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 6 & 7 & 1 \end{bmatrix},\ B = \begin{bmatrix} 1 & 3 \\ 1 & 1 \\ 3 & 1 \end{bmatrix}.$$

(a) Calculate *AB* and *BA*.

(4 marks)

(b) Show that $(AB)^T = B^T A^T$.

(2 marks)

(c) Comment on the singularity of AB and BA.

(2 marks)

(d) Explain why the |AB| not equal to |A||B|.

(2 marks)

3. Let x be an index of the total amount of goods consumed in an economy and z be a measure of the level of pollution. If $u(x,z) = 5x + \frac{\sqrt{2x}}{z}$ measures the total well-being of the economy. Find all first order and second order partial derivatives and interpret the economic meaning of their signs.

(10 marks)

4. Suppose that a monopolist faces the demand function

$$Q = (10 - P)^2$$

and has the cost function

$$C(Q) = Q^{\frac{3}{2}} + 4Q + 4$$

where *P* is price, *Q* is quantity, and 0 ≤ *P* ≤ *10 and 0* ≤ *Q* ≤ 10.

(a) Suppose that the monopolist maximises profit. Find *Q* that solves the monopolist's profit optimisation problem and find the corresponding price and profit.

(5 marks)

(b) Find the price elasticity of demand at the price obtained in (a) and find the consumer surplus at this price.

(5 marks)

5. A firm's output is predetermined at *10* units and it wishes to minimize the total cost of producing this level of output. The firm's inputs are capital, *K*, and labour, *L*. Its production function is

$$Q(K, L) = 2K^{\frac{2}{3}}L^{\frac{1}{3}}$$

The unit costs of *K* and *L* are *2* and *1* respectively.

(a) What is the Lagrangean function for this firm's optimization problem? (5 marks)

(b) Obtain the first order conditions and show the firm's demands for the input.

(15 marks)

(c) Confirm that this solution generates a minimum of cost subject to the output constraint and work out the minimum cost.

(10 marks)

6. Consider the matrix

$$A = \begin{bmatrix} 8 & 20 \\ 4 & 6 \end{bmatrix}.$$

(a) Compute the eigenvalues of A.

(10 marks)

(b) Compute the associated normalised eigenvectors.

(10 marks)

(c) Use the eigenvalues to calculate the trace and determinant of this matrix. Verify these results by calculating the trace and determinant by other means.

(10 marks)

(End of question paper)